

DHRUV GHOSAL

AI/ML Research Assistant · AIoT & Embedded Systems · Published Researcher

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SUMMARY

Computer Science undergraduate with four published research papers spanning AIoT, federated learning, differential privacy, and vision-transformer forensics. Combines hands-on ML/RAG engineering with embedded-systems and applied research experience — comfortable owning a problem from literature review through implementation and evaluation. Available for part-time, evening/night remote work.

RESEARCH & PUBLICATIONS

- **Optimizing Hydroponics and Aeroponics through AIoT: Future Perspectives in Controlled-Environment Farming** — multi-sensor arrays, automation algorithms, computer vision, solar backup, data encryption.
- **Differentially Private Blockchain-Based Climate Forecasting Framework** — DP-SGD-trained LSTM with attention + SHA-256 blockchain hashing for tamper-proof storage; RMSE 0.0774, MAE 0.0638, R^2 0.688.
- **Blockchain-Enabled Federated Learning Framework for Data Integrity in Digital Metrology (BFL-Metrology)** — 99.6% blockchain integrity verification; accuracy improved 91% → 94%.
- **Detecting AI-Generated and Altered Images Using Vision-Transformer-Based Forensic Learning** — ViT with FFT/DCT frequency-domain analysis; near-perfect AUC of 0.98.

EXPERIENCE

IT/Software Department Intern — Airports Authority of India (AAI), Northern Regional HQ

Jun 22 – Jul 23, 2026

- Pitched and built an AI NOTAM Assistant — a Retrieval-Augmented Generation (RAG)-based chatbot enabling plain-English queries over live NOTAM (Notice to Air Missions) data fetched via the United States Federal Aviation Administration (FAA) API, FastAPI, Pinecone, and Next.js.
- Produced a pitch document and 6-week delivery roadmap that was mentor-approved for continued development.

IoT Intern — ElevanceSkills

May 2026 – Jun 2026 · Remote

- Built and simulated an ESP32-based patient health-monitoring system on FreeRTOS with MQTT telemetry to Adafruit IO.
- Implemented dual real-time dashboards, automated medication alerts, servo-driven bed elevation, adaptive sensor sampling, and offline data buffering with reconnect sync.
- Owned the full project lifecycle from prototyping to simulation; awarded a Certificate of Completion.

ML / AI PROJECTS

AI NOTAM Assistant — AAI Internship Project

- Built an AI-powered RAG chatbot enabling plain-English queries over live NOTAM (Notice to Air Missions) data, using the FAA API for retrieval with FastAPI, Pinecone, and Next.js.
- Prepared a pitch document and mentor-approved 6-week delivery roadmap for continued development.

AI Resume Chatbot — RAG System — Personal Project

- Built a Retrieval-Augmented Generation chatbot using locally hosted LLMs (Ollama — Llama 3.1 / Qwen 2.5), LangChain, and ChromaDB for privacy-preserving inference.
- Processed PDF/DOCX resumes into semantic embeddings (nomic-embed-text) in ChromaDB; served retrieval-plus-generation via Flask REST APIs with multi-turn chat history.

Agentic AI Note-Taking Assistant — Personal Project

- Developed an agent-based assistant with a RAG pipeline (FAISS / ChromaDB) supporting multi-step reasoning and autonomous tool use.

AIoT Hydroponics / Aeroponics System — Personal Project

- Designed an AIoT architecture with multi-sensor arrays for real-time environmental monitoring and automation of soilless farming.
- Integrated computer vision for plant-health monitoring and ML-based crop optimization; formalized the framework in a published research paper.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, HTML/CSS

ML / AI: RAG pipelines, LangChain, ChromaDB / FAISS, Ollama, PyTorch-based experimentation, differential privacy (DP-SGD), federated learning, computer vision

Embedded / IoT: ESP32, Arduino IDE, Raspberry Pi, IoT sensors, SPIFFS, FreeRTOS, MQTT

Tools: GitHub, VS Code, Adafruit IO, NotebookLM, Antigravity

EDUCATION

B.Tech in Computer Science Engineering | 2024 – 2028

Maharaja Surajmal Institute of Technology (MSIT), Delhi

LEADERSHIP & EXTRACURRICULAR

- **Hardware / Robotics Lead — Melerigamy Teach Group:** led hardware and robotics projects; mentored students on sensor integration, embedded systems, and IoT best practices.

- **Project Committee, Hardware — IEEE MSIT:** contributed to hardware-focused projects bridging embedded systems and real-world engineering.
- **Hardware Member — Eyantra, MSIT (Aug 2024 – Sep 2025):** built hardware projects in a competitive robotics environment, applying embedded systems and Python.